# Evaluation runs: 2pc (a1-commit)

generated 2026-11-08 from brain-data 522afaee3 2026-11-08 12:20:57 +0100

Every campaign run on this benchmark, one row per instance (variant, specification, communication model, size) and engine; the newest run of an instance wins. Expected verdicts (the rule file): correct: expected conforms (n=1 is refused: the program never spawns visible p1); control-reverse: expected conforms (n=1 is refused: the program never spawns visible p1); eager: expected violation (n=1 is refused: the program never spawns visible p1); early-abort: expected violation (n=1 is refused: the program never spawns visible p1); silent: expected violation (n=1 is refused: the program never spawns visible p1). Blue rows are at the current pins (TraceForge 367cdc6, whose predecessor 9a0629e is equivalent for every engine but the join; merge join 367cdc6; JMC 3c43b2d2): a rerun gives the same result. Red rows are superseded (an older pin, or the in-memory join that the merge join replaces) and their rerun is pending. A planned instance without a measurement is not a row: it is listed under "Not run". $G_{ref}$, $G_{cand}$ = the graphs of the reference and of the implementation (the stateless engine's own count of specification graphs explored, repeats included, is $G_{ref}^{expl}$); $G_x$ = the implementation graphs no specification graph covers (violating pairs); T = wall seconds; M = peak resident MiB; tier = the run's time and memory limits; * marks a verdict other than the expected one; 🕓 = the time tier was reached; oom = the memory limit was reached.

## stateless

### stateless: pair **control-reverse**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | sweeps | $G_{ref}^{expl}$ | run |
|---|---|---|---|---|---|---|---|---|---|
| 2,1 | 16 | 8 | conforms | 0.0 | 4 | 1 h, 250 GB | 4 | 29 | brain03, 2026-11-03 |
| 2,2 | 256 | 64 | conforms | 0.1 | 4 | 1 h, 250 GB | 16 | 1658 | brain03, 2026-11-03 |
| 3,1 | 16 | 48 | conforms | 0.0 | 4 | 1 h, 250 GB | 5 | 38 | brain03, 2026-11-03 |
| 4,2 | 256 | 147456 | conforms | 4.2 | 6 | 1 h, 250 GB | 25 | 2353 | brain03, 2026-11-03 |
| 5,3 | | | 🕓 (1 h) | | | 1 h, 250 GB | | | brain03, 2026-11-03 |

### stateless: pair **correct**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | sweeps | $G_{ref}^{expl}$ | run |
|---|---|---|---|---|---|---|---|---|---|
| 2,1 | 16 | 8 | conforms | 0.0 | 4 | 1 h, 250 GB | 4 | 30 | brain03, 2026-11-03 |
| 2,2 | 256 | 64 | conforms | 0.1 | 4 | 1 h, 250 GB | 16 | 1847 | brain03, 2026-11-03 |
| 3,1 | 16 | 48 | conforms | 0.0 | 4 | 1 h, 250 GB | 5 | 30 | brain03, 2026-11-03 |
| 4,2 | 256 | 147456 | conforms | 4.2 | 2 | 48 h, 250 GB | 26 | 3177 | brain05, 2026-11-07 |
| 5,3 | | | 🕓 (1 h) | | | 1 h, 250 GB | | | brain03, 2026-11-03 |

### stateless: pair **eager**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | sweeps | $G_{ref}^{expl}$ | run |
|---|---|---|---|---|---|---|---|---|---|---|
| 2,1 | 16 | 8 | 5 | violation | 0.0 | 4 | 1 h, 250 GB | 8 | 101 | brain03, 2026-11-03 |
| 2,2 | 256 | 64 | 55 | violation | 0.7 | 2 | 1 h, 250 GB | 64 | 14966 | brain03, 2026-11-03 |
| 3,1 | 16 | 48 | 42 | violation | 0.1 | 4 | 1 h, 250 GB | 45 | 698 | brain03, 2026-11-03 |
| 4,2 | 256 | 147456 | 146880 | violation | 502.1 | 19 | 1 h, 250 GB | 146889 | 37601984 | brain03, 2026-11-03 |
| 5,3 | | | | 🕓 (1 h) | | | 1 h, 250 GB | | | brain03, 2026-11-03 |

### stateless: pair **early-abort**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | sweeps | $G_{ref}^{expl}$ | run |
|---|---|---|---|---|---|---|---|---|---|---|
| 2,1 | 16 | 8 | 4 | violation | 0.0 | 4 | 1 h, 250 GB | 7 | 83 | brain03, 2026-11-03 |
| 2,2 | 256 | 64 | 48 | violation | 0.6 | 4 | 1 h, 250 GB | 57 | 13455 | brain03, 2026-11-03 |

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | sweeps | $G_{ref}^{expl}$ | run |
|---|---|---|---|---|---|---|---|---|---|---|
| 3,1 | 16 | 48 | 32 | violation | 0.1 | 4 | 1 h, 250 GB | 36 | 545 | brain03, 2026-11-03 |
| 4,2 | 256 | 147456 | 139712 | violation | 390.6 | 19 | 48 h, 250 GB | 139728 | 35767283 | brain05, 2026-11-07 |
| 5,3 | | | | (1 h) | | | 1 h, 250 GB | | | brain03, 2026-11-03 |

### stateless: pair **silent**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | sweeps | $G_{ref}^{expl}$ | run |
|---|---|---|---|---|---|---|---|---|---|---|
| 2,1 | 16 | 8 | 8 | violation | 0.0 | 4 | 1 h, 250 GB | 8 | 128 | brain03, 2026-11-03 |
| 2,2 | 256 | 64 | 64 | violation | 0.8 | 4 | 1 h, 250 GB | 64 | 16384 | brain03, 2026-11-03 |
| 3,1 | 16 | 48 | 48 | violation | 0.1 | 4 | 1 h, 250 GB | 48 | 768 | brain03, 2026-11-03 |
| 4,2 | 256 | 147456 | 147456 | violation | 482.9 | 19 | 1 h, 250 GB | 147456 | 37748736 | brain03, 2026-11-03 |
| 5,3 | | | | (1 h) | | | 1 h, 250 GB | | | brain03, 2026-11-03 |

**Not run (stateless).** planned but never scheduled (grid points beyond the ladder): eager 2 3 fifo, control-reverse 2 3 fifo, correct 2 3 fifo, silent 2 3 fifo, early-abort 2 3 fifo, early-abort 2 4 fifo, control-reverse 2 4 fifo, correct 2 4 fifo, silent 2 4 fifo, eager 2 4 fifo, early-abort 2 5 fifo, control-reverse 2 5 fifo, silent 2 5 fifo, correct 2 5 fifo, eager 2 5 fifo, silent 3 2 fifo, control-reverse 3 2 fifo, correct 3 2 fifo, early-abort 3 2 fifo, eager 3 2 fifo, correct 3 3 fifo, eager 3 3 fifo, control-reverse 3 3 fifo, early-abort 3 3 fifo, … (101 instances in all).

left out by the table rules (ok): early-abort 4 2 fifo, correct 4 2 fifo.

skipped: a smaller size had reached its limit: silent 6 4 fifo, eager 6 4 fifo, control-reverse 6 4 fifo, correct 6 4 fifo, early-abort 6 4 fifo, early-abort 7 5 fifo, correct 7 5 fifo, control-reverse 7 5 fifo, silent 7 5 fifo, eager 7 5 fifo, silent 8 6 fifo, early-abort 8 6 fifo, control-reverse 8 6 fifo, eager 8 6 fifo, correct 8 6 fifo, early-abort 9 7 fifo, control-reverse 9 7 fifo, eager 9 7 fifo, correct 9 7 fifo, silent 9 7 fifo.

## stateful

### stateful: pair **control-reverse**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | run |
|---|---|---|---|---|---|---|---|
| 2,1 | 16 | 8 | conforms | 0.3 | 2 | 1 h, 250 GB | brain02, 2026-11-03 |
| 2,2 | 256 | 64 | conforms | 0.3 | 2 | 1 h, 250 GB | brain02, 2026-11-03 |
| 3,1 | 16 | 48 | conforms | 0.3 | 2 | 1 h, 250 GB | brain02, 2026-11-03 |
| 4,2 | 256 | 147456 | conforms | 4.7 | 2 | 1 h, 250 GB | brain02, 2026-11-03 |
| 5,3 | | | (1 h) | | | 1 h, 250 GB | brain02, 2026-11-03 |

### stateful: pair **correct**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | run |
|---|---|---|---|---|---|---|---|
| 2,1 | 16 | 8 | conforms | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-03 |
| 2,2 | 256 | 64 | conforms | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-03 |
| 3,1 | 16 | 48 | conforms | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-03 |
| 4,2 | 256 | 147456 | conforms | 4.1 | 2 | 48 h, 250 GB | brain05, 2026-11-07 |
| 5,3 | | | (1 h) | | | 1 h, 250 GB | brain02, 2026-11-03 |

### stateful: pair **eager**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | run |
|---|---|---|---|---|---|---|---|---|
| 2,1 | 16 | 8 | 5 | violation | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-03 |
| 2,2 | 256 | 64 | 55 | violation | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-03 |
| 3,1 | 16 | 48 | 42 | violation | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-03 |
| 4,2 | 256 | 147456 | 146880 | violation | 5.1 | 8 | 1 h, 250 GB | brain02, 2026-11-03 |
| 5,3 | | | | (1 h) | | | 1 h, 250 GB | brain02, 2026-11-03 |

### stateful: pair **early-abort**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | run |
|---|---|---|---|---|---|---|---|---|
| 2,1 | 16 | 8 | 4 | violation | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-03 |
| 2,2 | 256 | 64 | 48 | violation | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-03 |
| 3,1 | 16 | 48 | 32 | violation | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-03 |
| 4,2 | 256 | 147456 | 139712 | violation | 3.7 | 2 | 48 h, 250 GB | brain05, 2026-11-07 |
| 5,3 | | | | (1 h) | | | 1 h, 250 GB | brain02, 2026-11-03 |

### stateful: pair **silent**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | run |
|---|---|---|---|---|---|---|---|---|
| 2,1 | 16 | 8 | 8 | violation | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-03 |
| 2,2 | 256 | 64 | 64 | violation | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-03 |
| 3,1 | 16 | 48 | 48 | violation | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-03 |
| 4,2 | 256 | 147456 | 147456 | violation | 4.5 | 4 | 1 h, 250 GB | brain02, 2026-11-03 |
| 5,3 | | | | (1 h) | | | 1 h, 250 GB | brain02, 2026-11-03 |

**Not run (stateful).** planned but never scheduled (grid points beyond the ladder): eager 2 3 fifo, control-reverse 2 3 fifo, correct 2 3 fifo, silent 2 3 fifo, early-abort 2 3 fifo, early-abort 2 4 fifo, control-reverse 2 4 fifo, correct 2 4 fifo, silent 2 4 fifo, eager 2 4 fifo, early-abort 2 5 fifo, control-reverse 2 5 fifo, silent 2 5 fifo, correct 2 5 fifo, eager 2 5 fifo, silent 3 2 fifo, control-reverse 3 2 fifo, correct 3 2 fifo, early-abort 3 2 fifo, eager 3 2 fifo, correct 3 3 fifo, eager 3 3 fifo, control-reverse 3 3 fifo, early-abort 3 3 fifo, … (101 instances in all).

left out by the table rules (ok): early-abort 4 2 fifo, correct 4 2 fifo.

skipped: a smaller size had reached its limit: silent 6 4 fifo, eager 6 4 fifo, control-reverse 6 4 fifo, correct 6 4 fifo, early-abort 6 4 fifo, early-abort 7 5 fifo, correct 7 5 fifo, control-reverse 7 5 fifo, silent 7 5 fifo, eager 7 5 fifo, silent 8 6 fifo, early-abort 8 6 fifo, control-reverse 8 6 fifo, eager 8 6 fifo, correct 8 6 fifo, early-abort 9 7 fifo, control-reverse 9 7 fifo, eager 9 7 fifo, correct 9 7 fifo, silent 9 7 fifo.

## offline

### offline: pair **control-reverse**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,1 | 16 | 8 | conforms | 0.1 | 8 | 10 min, 16 GB | 0.0 | 0.1 | 0.0 | in-memory | brain05, 2026-11-02 |
| 2,2 | 256 | 64 | conforms | 0.1 | 8 | 10 min, 16 GB | 0.1 | 0.1 | 0.0 | in-memory | brain05, 2026-11-02 |
| 3,1 | 16 | 48 | conforms | 0.1 | 8 | 10 min, 16 GB | 0.1 | 0.1 | 0.0 | in-memory | brain05, 2026-11-02 |
| 4,2 | 256 | 147456 | conforms | 5.2 | 8 | 4 h, 250 GB | 0.1 | 5.2 | 0.0 | merge | brain01, 2026-11-03 |

### offline: pair **correct**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,1 | 16 | 8 | conforms | 0.1 | 8 | 10 min, 16 GB | 0.0 | 0.1 | 0.0 | in-memory | brain05, 2026-11-02 |
| 2,2 | 256 | 64 | conforms | 0.2 | 8 | 10 min, 16 GB | 0.1 | 0.2 | 0.0 | in-memory | brain05, 2026-11-02 |
| 3,1 | 16 | 48 | conforms | 0.1 | 8 | 1 h, 250 GB | 0.1 | 0.1 | 0.0 | merge | brain05, 2026-11-03 |
| 4,2 | 256 | 147456 | conforms | 5.9 | 8 | 4 h, 250 GB | 0.1 | 5.9 | 0.0 | merge | brain01, 2026-11-03 |
| 4,2 | 256 | 147456 | conforms | 4.3 | 6 | 48 h, 250 GB | 0.1 | 4.3 | 0.0 | merge | brain05, 2026-11-07 |

### offline: pair **eager**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,1 | 16 | 8 | 5 | violation | 0.1 | 8 | 10 min, 16 GB | 0.0 | 0.1 | 0.0 | in-memory | brain05, 2026-11-02 |
| 2,2 | 256 | 64 | 55 | violation | 0.2 | 8 | 10 min, 16 GB | 0.1 | 0.2 | 0.0 | in-memory | brain05, 2026-11-02 |
| 3,1 | 16 | 48 | 25 | violation | 0.1 | 8 | 1 h, 250 GB | 0.1 | 0.1 | 0.0 | merge | brain05, 2026-11-03 |
| 4,2 | 256 | 147456 | 1015 | violation | 5.7 | 10 | 4 h, 250 GB | 0.1 | 5.7 | 0.0 | merge | brain01, 2026-11-05 |

### offline: pair **early-abort**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,1 | 16 | 8 | 4 | violation | 0.1 | 8 | 10 min, 16 GB | 0.0 | 0.1 | 0.0 | in-memory | brain05, 2026-11-02 |
| 2,2 | 256 | 64 | 40 | violation | 0.1 | 8 | 10 min, 16 GB | 0.1 | 0.1 | 0.0 | in-memory | brain05, 2026-11-02 |
| 3,1 | 16 | 48 | 12 | violation | 0.1 | 8 | 10 min, 16 GB | 0.1 | 0.1 | 0.0 | in-memory | brain05, 2026-11-02 |
| 4,2 | 256 | 147456 | 240 | violation | 5.2 | 8 | 4 h, 250 GB | 0.1 | 5.1 | 0.0 | merge | brain01, 2026-11-05 |
| 4,2 | 256 | 147456 | 240 | violation | 3.8 | 6 | 48 h, 250 GB | 0.1 | 3.8 | 0.0 | merge | brain05, 2026-11-07 |

### offline: pair **silent**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,1 | 16 | 8 | 4 | violation | 0.1 | 8 | 10 min, 16 GB | 0.0 | 0.1 | 0.0 | in-memory | brain05, 2026-11-02 |
| 2,2 | 256 | 64 | 16 | violation | 0.1 | 8 | 10 min, 16 GB | 0.1 | 0.1 | 0.0 | in-memory | brain05, 2026-11-02 |
| 3,1 | 16 | 48 | 5 | violation | 0.1 | 8 | 10 min, 16 GB | 0.1 | 0.1 | 0.0 | in-memory | brain05, 2026-11-02 |
| 4,2 | 256 | 147456 | 25 | violation | 7.2 | 10 | 4 h, 250 GB | 0.1 | 7.2 | 0.0 | merge | brain01, 2026-11-05 |

**Not run (offline).** planned but never scheduled (grid points beyond the ladder): eager 2 3 fifo, control-reverse 2 3 fifo, correct 2 3 fifo, silent 2 3 fifo, early-abort 2 3 fifo, early-abort 2 4 fifo, control-reverse 2 4 fifo, correct 2 4 fifo, silent 2 4 fifo, eager 2 4 fifo, early-abort 2 5 fifo, control-reverse 2 5 fifo, silent 2 5 fifo, correct 2 5 fifo, eager 2 5 fifo, silent 3 2 fifo, control-reverse 3 2 fifo, correct 3 2 fifo, early-abort 3 2 fifo, eager 3 2 fifo, correct 3 3 fifo, eager 3 3 fifo, control-reverse 3 3 fifo, early-abort 3 3 fifo, … (106 instances in all).

skipped: a smaller size had reached its limit: control-reverse 5 3 fifo, early-abort 5 3 fifo, eager 5 3 fifo, silent 5 3 fifo, correct 5 3 fifo, silent 6 4 fifo, eager 6 4 fifo, correct 6 4 fifo, control-reverse 6 4 fifo, early-abort 6 4 fifo, early-abort 7 5 fifo, correct 7 5 fifo, control-reverse 7 5 fifo, silent 7 5 fifo, eager 7 5 fifo, silent 8 6 fifo, early-abort 8 6 fifo, control-reverse 8 6 fifo, eager 8 6 fifo, correct 8 6 fifo, early-abort 9 7 fifo, control-reverse 9 7 fifo, eager 9 7 fifo, correct 9 7 fifo, … (25 instances in all).

## cross offline, Java implementation against the Rust specification

### cross offline, Java implementation against the Rust specification: pair **control-reverse**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,1 | 16 | 8 | conforms | 3.1 | 179 | 10 min, 16 GB | 0.0 | 3.1 | 0.0 | in-memory | brain01, 2026-11-01 |
| 2,2 | 256 | 64 | conforms | 3.6 | 170 | 10 min, 16 GB | 0.1 | 3.6 | 0.0 | in-memory | brain01, 2026-11-01 |
| 3,1 | 16 | 48 | conforms | 3.3 | 168 | 10 min, 16 GB | 0.1 | 3.3 | 0.0 | in-memory | brain01, 2026-11-01 |
| 3,2 | 256 | 2304 | conforms | 10.4 | 2321 | 48 h, 80 GB | 0.2 | 10.4 | 0.0 | merge | pinky12, 2026-11-07 |

### cross offline, Java implementation against the Rust specification: pair **correct**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,1 | 16 | 8 | conforms | 3.1 | 166 | 10 min, 16 GB | 0.0 | 3.1 | 0.0 | in-memory | brain01, 2026-11-01 |
| 2,2 | 256 | 64 | conforms | 3.4 | 180 | 10 min, 16 GB | 0.1 | 3.4 | 0.0 | in-memory | brain01, 2026-11-01 |
| 3,1 | 16 | 48 | conforms | 3.4 | 166 | 10 min, 16 GB | 0.1 | 3.4 | 0.0 | in-memory | brain01, 2026-11-01 |
| 3,2 | 256 | 2304 | conforms | 10.3 | 2395 | 48 h, 80 GB | 0.2 | 10.3 | 0.0 | merge | pinky12, 2026-11-07 |

### cross offline, Java implementation against the Rust specification: pair **eager**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,1 | 16 | 8 | 5 | violation | 3.1 | 160 | 10 min, 16 GB | 0.0 | 3.1 | 0.0 | in-memory | brain01, 2026-11-01 |
| 2,2 | 256 | 64 | 55 | violation | 3.4 | 164 | 10 min, 16 GB | 0.1 | 3.4 | 0.0 | in-memory | brain01, 2026-11-01 |
| 3,1 | 16 | 48 | 25 | violation | 3.3 | 171 | 10 min, 16 GB | 0.1 | 3.3 | 0.0 | in-memory | brain01, 2026-11-01 |
| 3,2 | 256 | 2304 | 775 | violation | 11.2 | 2584 | 48 h, 80 GB | 0.2 | 11.1 | 0.0 | merge | pinky12, 2026-11-07 |

### cross offline, Java implementation against the Rust specification: pair **early-abort**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,1 | 16 | 8 | 4 | violation | 3.2 | 178 | 10 min, 16 GB | 0.0 | 3.2 | 0.0 | in-memory | brain01, 2026-11-01 |
| 2,2 | 256 | 64 | 40 | violation | 3.4 | 179 | 10 min, 16 GB | 0.1 | 3.4 | 0.0 | in-memory | brain01, 2026-11-01 |
| 3,1 | 16 | 48 | 12 | violation | 3.4 | 172 | 10 min, 16 GB | 0.1 | 3.4 | 0.0 | in-memory | brain01, 2026-11-01 |
| 3,2 | 256 | 2304 | 240 | violation | 10.7 | 2338 | 48 h, 80 GB | 0.2 | 10.7 | 0.0 | merge | pinky12, 2026-11-07 |

### cross offline, Java implementation against the Rust specification: pair **silent**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,1 | 16 | 8 | 4 | violation | 3.1 | 156 | 10 min, 16 GB | 0.0 | 3.1 | 0.0 | in-memory | brain01, 2026-11-01 |
| 2,2 | 256 | 64 | 16 | violation | 3.4 | 174 | 10 min, 16 GB | 0.1 | 3.4 | 0.0 | in-memory | brain01, 2026-11-01 |
| 3,1 | 16 | 48 | 5 | violation | 3.2 | 171 | 10 min, 16 GB | 0.1 | 3.2 | 0.0 | in-memory | brain01, 2026-11-01 |
| 3,2 | 256 | 2304 | 25 | violation | 10.4 | 2372 | 48 h, 80 GB | 0.2 | 10.4 | 0.0 | merge | pinky12, 2026-11-07 |

**Not run (cross offline, Java implementation against the Rust specification).** planned but never scheduled (grid points beyond the ladder): eager 2 3 fifo, early-abort 2 3 fifo, correct 2 3 fifo, control-reverse 2 3 fifo, silent 2 3 fifo, silent 4 1 fifo, eager 4 1 fifo, control-reverse 4 1 fifo, correct 4 1 fifo, early-abort 4 1 fifo.

## cross offline, Rust implementation against the Java specification

### cross offline, Rust implementation against the Java specification: pair **control-reverse**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,1 | 16 | 8 | conforms | 3.2 | 178 | 10 min, 16 GB | 3.2 | 0.1 | 0.0 | in-memory | brain01, 2026-11-01 |
| 2,2 | 256 | 64 | conforms | 4.0 | 178 | 10 min, 16 GB | 4.0 | 0.1 | 0.0 | in-memory | brain01, 2026-11-01 |
| 2,3 | 4096 | 512 | conforms | 10.6 | 2971 | 48 h, 200 GB | 10.6 | 0.1 | 0.0 | merge | volta14, 2026-11-07 |
| 3,1 | 16 | 48 | conforms | 3.1 | 173 | 10 min, 16 GB | 3.1 | 0.1 | 0.0 | in-memory | brain01, 2026-11-01 |

### cross offline, Rust implementation against the Java specification: pair **correct**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,1 | 16 | 8 | conforms | 3.2 | 178 | 10 min, 16 GB | 3.2 | 0.1 | 0.0 | in-memory | brain01, 2026-11-01 |
| 2,2 | 256 | 64 | conforms | 4.0 | 178 | 10 min, 16 GB | 4.0 | 0.2 | 0.0 | in-memory | brain01, 2026-11-01 |
| 2,3 | 4096 | 512 | conforms | 10.6 | 2971 | 48 h, 200 GB | 10.6 | 0.1 | 0.0 | merge | volta14, 2026-11-07 |
| 3,1 | 16 | 48 | conforms | 3.1 | 173 | 10 min, 16 GB | 3.1 | 0.1 | 0.0 | in-memory | brain01, 2026-11-01 |

### cross offline, Rust implementation against the Java specification: pair **eager**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,1 | 16 | 8 | 5 | violation | 3.2 | 178 | 10 min, 16 GB | 3.2 | 0.1 | 0.0 | in-memory | brain01, 2026-11-01 |
| 2,2 | 256 | 64 | 55 | violation | 4.0 | 178 | 10 min, 16 GB | 4.0 | 0.2 | 0.0 | in-memory | brain01, 2026-11-01 |
| 2,3 | 4096 | 512 | 485 | violation | 10.6 | 2971 | 48 h, 200 GB | 10.6 | 0.1 | 0.0 | merge | volta14, 2026-11-07 |
| 3,1 | 16 | 48 | 25 | violation | 3.1 | 172 | 10 min, 16 GB | 3.1 | 0.1 | 0.0 | in-memory | brain01, 2026-11-01 |

### cross offline, Rust implementation against the Java specification: pair **early-abort**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,1 | 16 | 8 | 4 | violation | 3.2 | 178 | 10 min, 16 GB | 3.2 | 0.1 | 0.0 | in-memory | brain01, 2026-11-01 |
| 2,2 | 256 | 64 | 40 | violation | 4.0 | 178 | 10 min, 16 GB | 4.0 | 0.1 | 0.0 | in-memory | brain01, 2026-11-01 |
| 2,3 | 4096 | 512 | 316 | violation | 10.6 | 2971 | 48 h, 200 GB | 10.6 | 0.1 | 0.0 | merge | volta14, 2026-11-07 |
| 3,1 | 16 | 48 | 12 | violation | 3.1 | 173 | 10 min, 16 GB | 3.1 | 0.1 | 0.0 | in-memory | brain01, 2026-11-01 |

### cross offline, Rust implementation against the Java specification: pair **silent**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,1 | 16 | 8 | 4 | violation | 3.2 | 178 | 10 min, 16 GB | 3.2 | 0.1 | 0.0 | in-memory | brain01, 2026-11-01 |
| 2,2 | 256 | 64 | 16 | violation | 4.0 | 178 | 10 min, 16 GB | 4.0 | 0.1 | 0.0 | in-memory | brain01, 2026-11-01 |
| 3,1 | 16 | 48 | 5 | violation | 3.1 | 172 | 10 min, 16 GB | 3.1 | 0.1 | 0.0 | in-memory | brain01, 2026-11-01 |

**Not run (cross offline, Rust implementation against the Java specification).** planned but never scheduled (grid points beyond the ladder): silent 2 3 fifo, control-reverse 3 2 fifo, eager 3 2 fifo, early-abort 3 2 fifo, silent 3 2 fifo, correct 3 2 fifo, silent 4 1 fifo, eager 4 1 fifo, control-reverse 4 1 fifo, correct 4 1 fifo, early-abort 4 1 fifo.

## cross online, Java implementation (JMC) against the Rust specification export

### cross online, Java implementation (JMC) against the Rust specification export: pair **control-reverse**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2,1 | 16 | 8 | conforms | 0.7 | 149 | 1 h, 250 GB | | brain05, 2026-11-02 |
| 2,2 | 256 | 64 | conforms | 1.1 | 148 | 1 h, 250 GB | | brain05, 2026-11-02 |
| 3,1 | 16 | 48 | conforms | 0.8 | 146 | 1 h, 250 GB | | brain05, 2026-11-02 |

### cross online, Java implementation (JMC) against the Rust specification export: pair **correct**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2,1 | 16 | 8 | conforms | 0.7 | 167 | 1 h, 250 GB | | brain05, 2026-11-02 |
| 2,2 | 256 | 64 | conforms | 1.1 | 152 | 1 h, 250 GB | | brain05, 2026-11-02 |
| 3,1 | 16 | 48 | conforms | 0.8 | 148 | 1 h, 250 GB | | brain05, 2026-11-02 |

### cross online, Java implementation (JMC) against the Rust specification export: pair **eager**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|---|
| 2,1 | 16 | 8 | 5 | violation | 0.7 | 146 | 1 h, 250 GB | 46 | brain05, 2026-11-02 |
| 2,2 | 256 | 64 | 55 | violation | 1.2 | 156 | 1 h, 250 GB | 52 | brain05, 2026-11-02 |
| 3,1 | 16 | 48 | 42 | violation | 0.9 | 150 | 1 h, 250 GB | 47 | brain05, 2026-11-02 |

### cross online, Java implementation (JMC) against the Rust specification export: pair **early-abort**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|---|
| 2,1 | 16 | 8 | 4 | violation | 0.7 | 148 | 1 h, 250 GB | 88 | brain05, 2026-11-02 |
| 2,2 | 256 | 64 | 48 | violation | 1.1 | 156 | 1 h, 250 GB | 124 | brain05, 2026-11-02 |
| 3,1 | 16 | 48 | 32 | violation | 0.9 | 148 | 1 h, 250 GB | 121 | brain05, 2026-11-02 |

### cross online, Java implementation (JMC) against the Rust specification export: pair **silent**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|---|
| 2,1 | 16 | 8 | 8 | violation | 0.7 | 146 | 1 h, 250 GB | 41 | brain05, 2026-11-02 |
| 2,2 | 256 | 64 | 64 | violation | 1.2 | 156 | 1 h, 250 GB | 65 | brain05, 2026-11-02 |
| 3,1 | 16 | 48 | 48 | violation | 0.9 | 167 | 1 h, 250 GB | 53 | brain05, 2026-11-02 |

**Not run (cross online, Java implementation (JMC) against the Rust specification export).** planned but never scheduled (grid points beyond the ladder): correct 2 1 fifo, silent 2 1 fifo, control-reverse 2 1 fifo, early-abort 2 1 fifo, eager 2 1 fifo, control-reverse 2 2 fifo, silent 2 2 fifo, eager 2 2 fifo, correct 2 2 fifo, early-abort 2 2 fifo, early-abort 3 1 fifo, correct 3 1 fifo, silent 3 1 fifo, eager 3 1 fifo, control-reverse 3 1 fifo.

## cross online, Rust implementation (TraceForge) against the Java specification export

### cross online, Rust implementation (TraceForge) against the Java specification export: pair **control-reverse**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2,1 | 16 | 8 | conforms | 3.2 | 174 | 1 h, 250 GB | | brain04, 2026-11-01 |
| 2,2 | 256 | 64 | conforms | 4.0 | 174 | 1 h, 250 GB | | brain04, 2026-11-01 |
| 3,1 | 16 | 48 | conforms | 3.2 | 168 | 1 h, 250 GB | | brain04, 2026-11-01 |

### cross online, Rust implementation (TraceForge) against the Java specification export: pair **correct**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2,1 | 16 | 8 | conforms | 3.2 | 174 | 1 h, 250 GB | | brain04, 2026-11-01 |
| 2,2 | 256 | 64 | conforms | 4.0 | 174 | 1 h, 250 GB | | brain04, 2026-11-01 |
| 3,1 | 16 | 48 | conforms | 3.2 | 168 | 1 h, 250 GB | | brain04, 2026-11-01 |

### cross online, Rust implementation (TraceForge) against the Java specification export: pair **eager**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|---|
| 2,1 | 16 | 8 | 5 | violation | 3.2 | 174 | 1 h, 250 GB | 1 | brain04, 2026-11-01 |
| 2,2 | 256 | 64 | 55 | violation | 4.0 | 174 | 1 h, 250 GB | 2 | brain04, 2026-11-01 |
| 3,1 | 16 | 48 | 42 | violation | 3.2 | 168 | 1 h, 250 GB | 1 | brain04, 2026-11-01 |

### cross online, Rust implementation (TraceForge) against the Java specification export: pair **early-abort**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|---|
| 2,1 | 16 | 8 | 4 | violation | 3.2 | 174 | 1 h, 250 GB | 1 | brain04, 2026-11-01 |
| 2,2 | 256 | 64 | 48 | violation | 4.0 | 174 | 1 h, 250 GB | 2 | brain04, 2026-11-01 |
| 3,1 | 16 | 48 | 32 | violation | 3.2 | 168 | 1 h, 250 GB | 2 | brain04, 2026-11-01 |

### cross online, Rust implementation (TraceForge) against the Java specification export: pair **silent**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|---|
| 2,1 | 16 | 8 | 8 | violation | 3.2 | 174 | 1 h, 250 GB | 2 | brain04, 2026-11-01 |
| 2,2 | 256 | 64 | 64 | violation | 4.0 | 174 | 1 h, 250 GB | 2 | brain04, 2026-11-01 |
| 3,1 | 16 | 48 | 48 | violation | 3.2 | 168 | 1 h, 250 GB | 2 | brain04, 2026-11-01 |

**Not run (cross online, Rust implementation (TraceForge) against the Java specification export).** left out by the table rules (ok): correct 2 1 fifo, silent 2 1 fifo, control-reverse 2 1 fifo, early-abort 2 1 fifo, eager 2 1 fifo, control-reverse 2 2 fifo, silent 2 2 fifo, eager 2 2 fifo, correct 2 2 fifo, early-abort 2 2 fifo, early-abort 3 1 fifo, correct 3 1 fifo, silent 3 1 fifo, eager 3 1 fifo, control-reverse 3 1 fifo.

# context-free twins offline, Rust implementation against the context-free Java port

## context-free twins offline, Rust implementation against the context-free Java port: pair control-reverse, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,3 | 512 | 512 | conforms* | 5.3 | 1523 | 48 h, 200 GB | 5.3 | 0.1 | 0.0 | merge | volta14, 2026-11-07 |
| 2,3 | 512 | 512 | conforms* | 5.3 | 210 | 48 h, 200 GB | 5.3 | 0.1 | 0.0 | merge | volta14, 2026-11-07 |

## context-free twins offline, Rust implementation against the context-free Java port: pair control-reverse, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,1 | 8 | 8 | conforms | 7.3 | 179 | 4 h, 250 GB | 7.3 | 0.5 | 0.0 | merge | brain01, 2026-11-03 |
| 2,2 | 64 | 64 | conforms | 11.8 | 1973 | 4 h, 250 GB | 11.8 | 0.1 | 0.0 | merge | brain01, 2026-11-03 |
| 3,1 | 48 | 48 | conforms | 8.8 | 153 | 4 h, 250 GB | 8.8 | 0.1 | 0.0 | merge | brain01, 2026-11-03 |

## context-free twins offline, Rust implementation against the context-free Java port: pair control-reverse, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,1 | 8 | 8 | conforms* | 11.1 | 1738 | 4 h, 250 GB | 11.1 | 0.5 | 0.0 | merge | brain01, 2026-11-03 |
| 2,2 | 64 | 64 | conforms* | 7.5 | 165 | 4 h, 250 GB | 7.5 | 0.1 | 0.0 | merge | brain01, 2026-11-03 |
| 3,1 | 48 | 48 | conforms* | 10.9 | 1813 | 4 h, 250 GB | 10.8 | 0.1 | 0.0 | merge | brain01, 2026-11-03 |

## context-free twins offline, Rust implementation against the context-free Java port: pair correct, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,3 | 512 | 512 | conforms* | 5.3 | 210 | 48 h, 200 GB | 5.3 | 0.1 | 0.0 | merge | volta14, 2026-11-07 |

## context-free twins offline, Rust implementation against the context-free Java port: pair correct, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,1 | 8 | 8 | conforms | 11.1 | 1738 | 4 h, 250 GB | 11.1 | 0.1 | 0.0 | merge | brain01, 2026-11-03 |
| 2,2 | 64 | 64 | conforms | 5.0 | 164 | 10 min, 250 GB | 4.8 | 0.2 | 0.1 | merge | brain02, 2026-11-03 |
| 3,1 | 48 | 48 | conforms | 4.9 | 179 | 10 min, 250 GB | 4.9 | 0.2 | 0.0 | merge | brain02, 2026-11-03 |

### context-free twins offline, Rust implementation against the context-free Java port: pair **eager**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,1 | 8 | 8 | conforms | 6.3 | 164 | 4 h, 250 GB | 6.3 | 0.1 | 0.0 | merge | brain01, 2026-11-03 |
| 2,2 | 64 | 64 | conforms | 4.3 | 161 | 10 min, 250 GB | 4.3 | 0.2 | 0.0 | merge | brain02, 2026-11-03 |
| 3,1 | 48 | 48 | conforms | 8.2 | 162 | 10 min, 250 GB | 8.2 | 0.2 | 0.0 | merge | brain02, 2026-11-03 |

### context-free twins offline, Rust implementation against the context-free Java port: pair **early-abort**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,1 | 8 | 8 | conforms | 9.8 | 161 | 4 h, 250 GB | 9.8 | 0.1 | 0.0 | merge | brain01, 2026-11-03 |
| 2,2 | 64 | 64 | conforms | 14.7 | 1568 | 4 h, 250 GB | 14.7 | 0.4 | 0.0 | merge | brain01, 2026-11-03 |
| 3,1 | 48 | 48 | conforms | 7.8 | 160 | 4 h, 250 GB | 7.8 | 0.5 | 0.0 | merge | brain01, 2026-11-03 |

### context-free twins offline, Rust implementation against the context-free Java port: pair **silent**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,1 | 8 | 8 | conforms | 12.4 | 1845 | 4 h, 250 GB | 12.4 | 0.5 | 0.0 | merge | brain01, 2026-11-03 |
| 2,2 | 64 | 64 | conforms | 15.2 | 1494 | 4 h, 250 GB | 15.2 | 0.1 | 0.0 | merge | brain01, 2026-11-03 |
| 3,1 | 48 | 48 | conforms | 4.9 | 162 | 4 h, 250 GB | 4.8 | 0.1 | 0.1 | merge | brain01, 2026-11-03 |

### context-free twins offline, Rust implementation against the context-free Java port: pair **eager**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,3 | 512 | 512 | | conforms* | 5.5 | 194 | 48 h, 200 GB | 5.5 | 0.1 | 0.0 | merge | volta14, 2026-11-07 |
| 2,3 | 512 | 512 | 485 | violation | 5.3 | 211 | 48 h, 200 GB | 5.3 | 0.1 | 0.0 | merge | volta14, 2026-11-07 |

### context-free twins offline, Rust implementation against the context-free Java port: pair **eager**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,1 | 8 | 8 | 5 | violation | 11.1 | 1738 | 4 h, 250 GB | 11.1 | 0.1 | 0.0 | merge | brain01, 2026-11-03 |
| 2,2 | 64 | 64 | 55 | violation | 4.8 | 164 | 10 min, 250 GB | 4.8 | 0.2 | 0.0 | merge | brain02, 2026-11-03 |
| 3,1 | 48 | 48 | 25 | violation | 4.9 | 179 | 10 min, 250 GB | 4.9 | 0.2 | 0.0 | merge | brain02, 2026-11-03 |

### context-free twins offline, Rust implementation against the context-free Java port: pair **early-abort**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,3 | 512 | 512 | | conforms* | 5.4 | 219 | 48 h, 200 GB | 5.4 | 0.1 | 0.0 | merge | volta14, 2026-11-07 |
| 2,3 | 512 | 512 | 316 | violation | 5.3 | 211 | 48 h, 200 GB | 5.3 | 0.1 | 0.0 | merge | volta14, 2026-11-07 |

### context-free twins offline, Rust implementation against the context-free Java port: pair **early-abort**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,1 | 8 | 8 | 4 | violation | 11.1 | 1738 | 4 h, 250 GB | 11.1 | 0.1 | 0.0 | merge | brain01, 2026-11-03 |

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,2 | 64 | 64 | 40 | violation | 7.5 | 165 | 4 h, 250 GB | 7.5 | 0.4 | 0.0 | merge | brain01, 2026-11-03 |
| 3,1 | 48 | 48 | 12 | violation | 10.8 | 1813 | 4 h, 250 GB | 10.8 | 0.5 | 0.0 | merge | brain01, 2026-11-03 |

### context-free twins offline, Rust implementation against the context-free Java port: pair **silent**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,3 | 512 | 512 |  | conforms* | 5.2 | 192 | 48 h, 200 GB | 5.2 | 0.1 | 0.0 | merge | volta14, 2026-11-07 |
| 2,3 | 512 | 512 | 64 | violation | 5.3 | 210 | 48 h, 200 GB | 5.3 | 0.1 | 0.0 | merge | volta14, 2026-11-07 |

### context-free twins offline, Rust implementation against the context-free Java port: pair **silent**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,1 | 8 | 8 | 4 | violation | 11.1 | 1738 | 4 h, 250 GB | 11.1 | 0.5 | 0.0 | merge | brain01, 2026-11-03 |
| 2,2 | 64 | 64 | 16 | violation | 7.5 | 165 | 4 h, 250 GB | 7.5 | 0.1 | 0.0 | merge | brain01, 2026-11-03 |
| 3,1 | 48 | 48 | 5 | violation | 10.8 | 1813 | 4 h, 250 GB | 10.8 | 0.1 | 0.0 | merge | brain01, 2026-11-03 |

**Not run (context-free twins offline, Rust implementation against the context-free Java port).** planned but never scheduled (grid points beyond the ladder): control-reverse 3 2 fifo, eager 3 2 fifo, early-abort 3 2 fifo, silent 3 2 fifo, correct 3 2 fifo, silent 4 1 fifo, eager 4 1 fifo, control-reverse 4 1 fifo, correct 4 1 fifo, early-abort 4 1 fifo.

## context-free twins offline, Java port against the Rust implementation

### context-free twins offline, Java port against the Rust implementation: pair **control-reverse**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,3 | 512 | 512 | conforms* | 5.3 | 1523 | 48 h, 200 GB | 0.1 | 5.3 | 0.0 | merge | volta14, 2026-11-07 |
| 2,3 | 512 | 512 | conforms* | 5.3 | 1523 | 48 h, 200 GB | 0.1 | 5.3 | 0.0 | merge | volta14, 2026-11-07 |

### context-free twins offline, Java port against the Rust implementation: pair **control-reverse**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,1 | 8 | 8 | conforms | 7.3 | 179 | 4 h, 250 GB | 0.5 | 7.3 | 0.0 | merge | brain01, 2026-11-03 |
| 2,2 | 64 | 64 | conforms | 11.8 | 1973 | 4 h, 250 GB | 0.1 | 11.8 | 0.0 | merge | brain01, 2026-11-05 |
| 3,1 | 48 | 48 | conforms | 8.8 | 153 | 4 h, 250 GB | 0.1 | 8.8 | 0.0 | merge | brain01, 2026-11-05 |

### context-free twins offline, Java port against the Rust implementation: pair **control-reverse**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,1 | 8 | 8 | conforms* | 7.3 | 179 | 4 h, 250 GB | 0.1 | 7.3 | 0.0 | merge | brain01, 2026-11-03 |
| 2,2 | 64 | 64 | conforms* | 11.8 | 1973 | 4 h, 250 GB | 0.2 | 11.8 | 0.0 | merge | brain01, 2026-11-05 |
| 3,1 | 48 | 48 | conforms* | 8.8 | 153 | 4 h, 250 GB | 0.2 | 8.8 | 0.0 | merge | brain01, 2026-11-05 |

### context-free twins offline, Java port against the Rust implementation: pair **correct**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,3 | 512 | 512 | conforms* | 5.3 | 210 | 48 h, 200 GB | 0.1 | 5.3 | 0.0 | merge | volta14, 2026-11-07 |

### context-free twins offline, Java port against the Rust implementation: pair **correct**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,1 | 8 | 8 | conforms | 11.1 | 1738 | 4 h, 250 GB | 0.1 | 11.1 | 0.0 | merge | brain01, 2026-11-05 |
| 2,2 | 64 | 64 | conforms | 4.8 | 164 | 10 min, 250 GB | 0.2 | 4.8 | 0.0 | merge | brain02, 2026-11-03 |
| 3,1 | 48 | 48 | conforms | 4.9 | 179 | 10 min, 250 GB | 0.2 | 4.9 | 0.0 | merge | brain02, 2026-11-03 |

### context-free twins offline, Java port against the Rust implementation: pair **eager**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,1 | 8 | 8 | conforms | 6.3 | 164 | 4 h, 250 GB | 0.1 | 6.3 | 0.0 | merge | brain01, 2026-11-05 |
| 2,2 | 64 | 64 | conforms | 4.3 | 161 | 10 min, 250 GB | 0.2 | 4.3 | 0.0 | merge | brain02, 2026-11-03 |
| 3,1 | 48 | 48 | conforms | 8.2 | 162 | 10 min, 250 GB | 0.2 | 8.2 | 0.0 | merge | brain02, 2026-11-03 |

### context-free twins offline, Java port against the Rust implementation: pair **early-abort**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,1 | 8 | 8 | conforms | 9.8 | 161 | 4 h, 250 GB | 0.1 | 9.8 | 0.0 | merge | brain01, 2026-11-05 |
| 2,2 | 64 | 64 | conforms | 14.7 | 1568 | 4 h, 250 GB | 0.4 | 14.7 | 0.0 | merge | brain01, 2026-11-05 |
| 3,1 | 48 | 48 | conforms | 7.8 | 160 | 4 h, 250 GB | 0.5 | 7.8 | 0.0 | merge | brain01, 2026-11-05 |

### context-free twins offline, Java port against the Rust implementation: pair **silent**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,1 | 8 | 8 | conforms | 12.5 | 1845 | 4 h, 250 GB | 0.5 | 12.4 | 0.0 | merge | brain01, 2026-11-05 |
| 2,2 | 64 | 64 | conforms | 15.2 | 1494 | 4 h, 250 GB | 0.1 | 15.2 | 0.0 | merge | brain01, 2026-11-05 |
| 3,1 | 48 | 48 | conforms | 4.8 | 162 | 4 h, 250 GB | 0.1 | 4.8 | 0.0 | merge | brain01, 2026-11-05 |

### context-free twins offline, Java port against the Rust implementation: pair **eager**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,3 | 512 | 512 | | conforms* | 5.5 | 194 | 48 h, 200 GB | 0.1 | 5.5 | 0.0 | merge | volta14, 2026-11-07 |
| 2,3 | 512 | 512 | 485 | violation | 5.5 | 194 | 48 h, 200 GB | 0.1 | 5.5 | 0.0 | merge | volta14, 2026-11-07 |

### context-free twins offline, Java port against the Rust implementation: pair **eager**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,1 | 8 | 8 | 5 | violation | 6.3 | 164 | 4 h, 250 GB | 0.1 | 6.3 | 0.0 | merge | brain01, 2026-11-05 |
| 2,2 | 64 | 64 | 55 | violation | 4.3 | 161 | 10 min, 250 GB | 0.2 | 4.3 | 0.0 | merge | brain02, 2026-11-03 |
| 3,1 | 48 | 48 | 25 | violation | 8.2 | 162 | 10 min, 250 GB | 0.2 | 8.2 | 0.0 | merge | brain02, 2026-11-03 |

### context-free twins offline, Java port against the Rust implementation: pair **early-abort**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,3 | 512 | 512 | | conforms* | 5.4 | 219 | 48 h, 200 GB | 0.1 | 5.4 | 0.0 | merge | volta14, 2026-11-07 |
| 2,3 | 512 | 512 | 316 | violation | 5.4 | 218 | 48 h, 200 GB | 0.1 | 5.4 | 0.0 | merge | volta14, 2026-11-07 |

### context-free twins offline, Java port against the Rust implementation: pair **early-abort**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,1 | 8 | 8 | 4 | violation | 9.8 | 161 | 4 h, 250 GB | 0.1 | 9.8 | 0.0 | merge | brain01, 2026-11-05 |
| 2,2 | 64 | 64 | 40 | violation | 14.7 | 1568 | 4 h, 250 GB | 0.2 | 14.7 | 0.0 | merge | brain01, 2026-11-05 |
| 3,1 | 48 | 48 | 12 | violation | 7.8 | 160 | 4 h, 250 GB | 0.2 | 7.8 | 0.0 | merge | brain01, 2026-11-05 |

### context-free twins offline, Java port against the Rust implementation: pair **silent**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,3 | 512 | 512 | 64 | violation | 5.2 | 192 | 48 h, 200 GB | 0.1 | 5.2 | 0.0 | merge | volta14, 2026-11-07 |
| 2,3 | 512 | 512 | | conforms* | 5.2 | 192 | 48 h, 200 GB | 0.1 | 5.2 | 0.0 | merge | volta14, 2026-11-07 |

### context-free twins offline, Java port against the Rust implementation: pair **silent**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,1 | 8 | 8 | 4 | violation | 12.4 | 1845 | 4 h, 250 GB | 0.1 | 12.4 | 0.0 | merge | brain01, 2026-11-05 |
| 2,2 | 64 | 64 | 16 | violation | 15.2 | 1494 | 4 h, 250 GB | 0.2 | 15.2 | 0.0 | merge | brain01, 2026-11-03 |
| 3,1 | 48 | 48 | 5 | violation | 4.8 | 162 | 4 h, 250 GB | 0.2 | 4.8 | 0.0 | merge | brain01, 2026-11-05 |

**Not run (context-free twins offline, Java port against the Rust implementation).** planned but never scheduled (grid points beyond the ladder): control-reverse 3 2 fifo, eager 3 2 fifo, early-abort 3 2 fifo, silent 3 2 fifo, correct 3 2 fifo, silent 4 1 fifo, eager 4 1 fifo, control-reverse 4 1 fifo, correct 4 1 fifo, early-abort 4 1 fifo.

## context-free twins online, Java port (JMC) against the Rust implementation's export

### context-free twins online, Java port (JMC) against the Rust implementation's export: pair **control-reverse**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2,1 | 8 | 8 | conforms | 3.8 | 164 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 2,2 | 64 | 64 | conforms | 3.9 | 162 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 3,1 | 48 | 48 | conforms | 3.6 | 158 | 1 h, 250 GB | | brain02, 2026-11-03 |

### context-free twins online, Java port (JMC) against the Rust implementation's export: pair **control-reverse**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2,1 | 8 | 8 | conforms* | 3.7 | 162 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 2,2 | 64 | 64 | conforms* | 3.9 | 167 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 3,1 | 48 | 48 | conforms* | 3.7 | 163 | 1 h, 250 GB | | brain02, 2026-11-03 |

**context-free twins online, Java port (JMC) against the Rust implementation's export: pair correct, model fifo, expected conforms**

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2,1 | 8 | 8 | conforms | 4.2 | 160 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 2,2 | 64 | 64 | conforms | 5.3 | 156 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 3,1 | 48 | 48 | conforms | 5.0 | 155 | 1 h, 250 GB | | brain02, 2026-11-03 |

**context-free twins online, Java port (JMC) against the Rust implementation's export: pair eager, model fifo, expected conforms**

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2,1 | 8 | 8 | conforms | 4.0 | 158 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 2,2 | 64 | 64 | conforms | 5.4 | 161 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 3,1 | 48 | 48 | conforms | 5.2 | 158 | 1 h, 250 GB | | brain02, 2026-11-03 |

**context-free twins online, Java port (JMC) against the Rust implementation's export: pair early-abort, model fifo, expected conforms**

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2,1 | 8 | 8 | conforms | 3.4 | 150 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 2,2 | 64 | 64 | conforms | 4.5 | 160 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 3,1 | 48 | 48 | conforms | 4.1 | 150 | 1 h, 250 GB | | brain02, 2026-11-03 |

**context-free twins online, Java port (JMC) against the Rust implementation's export: pair silent, model fifo, expected conforms**

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2,1 | 8 | 8 | conforms | 3.9 | 164 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 2,2 | 64 | 64 | conforms | 4.0 | 162 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 3,1 | 48 | 48 | conforms | 3.6 | 154 | 1 h, 250 GB | | brain02, 2026-11-03 |

**context-free twins online, Java port (JMC) against the Rust implementation's export: pair eager, model fifo, expected violation**

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|---|
| 2,1 | 8 | 8 | 5 | violation | 3.7 | 164 | 1 h, 250 GB | 132 | brain02, 2026-11-03 |
| 2,2 | 64 | 64 | 55 | violation | 5.2 | 155 | 1 h, 250 GB | 128 | brain02, 2026-11-03 |
| 3,1 | 48 | 48 | 42 | violation | 4.7 | 150 | 1 h, 250 GB | 86 | brain02, 2026-11-03 |

**context-free twins online, Java port (JMC) against the Rust implementation's export: pair early-abort, model fifo, expected violation**

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|---|
| 2,1 | 8 | 8 | 4 | violation | 3.7 | 164 | 1 h, 250 GB | 2260 | brain02, 2026-11-03 |
| 2,2 | 64 | 64 | 48 | violation | 4.0 | 157 | 1 h, 250 GB | 2340 | brain02, 2026-11-03 |
| 3,1 | 48 | 48 | 32 | violation | 4.0 | 158 | 1 h, 250 GB | 2196 | brain02, 2026-11-03 |

**context-free twins online, Java port (JMC) against the Rust implementation's export: pair silent, model fifo, expected violation**

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|---|
| 2,1 | 8 | 8 | 8 | violation | 3.7 | 166 | 1 h, 250 GB | 64 | brain02, 2026-11-03 |
| 2,2 | 64 | 64 | 64 | violation | 3.9 | 157 | 1 h, 250 GB | 85 | brain02, 2026-11-03 |
| 3,1 | 48 | 48 | 48 | violation | 4.0 | 153 | 1 h, 250 GB | 69 | brain02, 2026-11-03 |

## context-free twins online, Rust implementation (TraceForge) against the Java port's export

### context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **control-reverse**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2,1 | 8 | 8 | conforms | 3.1 | 154 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 2,2 | 64 | 64 | conforms | 3.6 | 167 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 3,1 | 48 | 48 | conforms | 3.8 | 160 | 1 h, 250 GB | | brain02, 2026-11-03 |

### context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **control-reverse**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2,1 | 8 | 8 | conforms* | 9.4 | 160 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 2,2 | 64 | 64 | conforms* | 3.6 | 164 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 3,1 | 48 | 48 | conforms* | 3.6 | 163 | 1 h, 250 GB | | brain02, 2026-11-03 |

### context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **correct**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2,1 | 8 | 8 | conforms | 9.6 | 160 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 2,2 | 64 | 64 | conforms | 3.6 | 164 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 3,1 | 48 | 48 | conforms | 3.6 | 163 | 1 h, 250 GB | | brain02, 2026-11-03 |

### context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **eager**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2,1 | 8 | 8 | conforms | 3.4 | 154 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 2,2 | 64 | 64 | conforms | 3.8 | 162 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 3,1 | 48 | 48 | conforms | 3.6 | 155 | 1 h, 250 GB | | brain02, 2026-11-03 |

### context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **early-abort**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2,1 | 8 | 8 | conforms | 3.3 | 160 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 2,2 | 64 | 64 | conforms | 3.6 | 154 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 3,1 | 48 | 48 | conforms | 3.4 | 164 | 1 h, 250 GB | | brain02, 2026-11-03 |

### context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **silent**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2,1 | 8 | 8 | conforms | 3.8 | 169 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 2,2 | 64 | 64 | conforms | 3.8 | 163 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 3,1 | 48 | 48 | conforms | 3.3 | 156 | 1 h, 250 GB | | brain02, 2026-11-03 |

**context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair eager, model fifo, expected violation**

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|---|
| 2,1 | 8 | 8 | 5 | violation | 9.4 | 160 | 1 h, 250 GB | 1 | brain02, 2026-11-03 |
| 2,2 | 64 | 64 | 55 | violation | 3.6 | 164 | 1 h, 250 GB | 1 | brain02, 2026-11-03 |
| 3,1 | 48 | 48 | 42 | violation | 3.6 | 163 | 1 h, 250 GB | 1 | brain02, 2026-11-03 |

**context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair early-abort, model fifo, expected violation**

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|---|
| 2,1 | 8 | 8 | 4 | violation | 9.4 | 160 | 1 h, 250 GB | 1 | brain02, 2026-11-03 |
| 2,2 | 64 | 64 | 48 | violation | 3.6 | 164 | 1 h, 250 GB | 2 | brain02, 2026-11-03 |
| 3,1 | 48 | 48 | 32 | violation | 3.6 | 163 | 1 h, 250 GB | 1 | brain02, 2026-11-03 |

**context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair silent, model fifo, expected violation**

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|---|
| 2,1 | 8 | 8 | 8 | violation | 9.4 | 160 | 1 h, 250 GB | 1 | brain02, 2026-11-03 |
| 2,2 | 64 | 64 | 64 | violation | 3.6 | 164 | 1 h, 250 GB | 1 | brain02, 2026-11-03 |
| 3,1 | 48 | 48 | 48 | violation | 3.6 | 163 | 1 h, 250 GB | 1 | brain02, 2026-11-03 |